



Understand the need, not just the keywords.

Semantic search & ranking for Vietnamese POIs — rigorously measured, explainable, and contract-compatible.

0.962

NDCG@5 · HELD-OUT TEST, EVALUATED ONCE

0.983

RECALL@3 · TEST

9.6_{ms}

P95 LATENCY (WARM)

• [demo link](#)

Vietnamese users search by **need** — not by place name

NEED

«mình đói bụng quá»

"I'm very hungry"

NO DIACRITICS

«quan cafe yen tinh»

"a quiet café"

TYP0

«yen tihn»

"quiet" (misspelled)

LANDMARK

«gần hồ gươm»

"near Hoàn Kiếm Lake"

KEYWORD · BM25 — «MINH DOI BUNG QUA, MINH O TPHCM» ("I'M VERY HUNGRY, I'M IN HCMC")

1 **Shopping mall** wrong category — ranked first

... **Gas station** appears in results

matches surface text; does not understand "hungry"

TASCO.TÌM · SEMANTIC — SAME QUERY

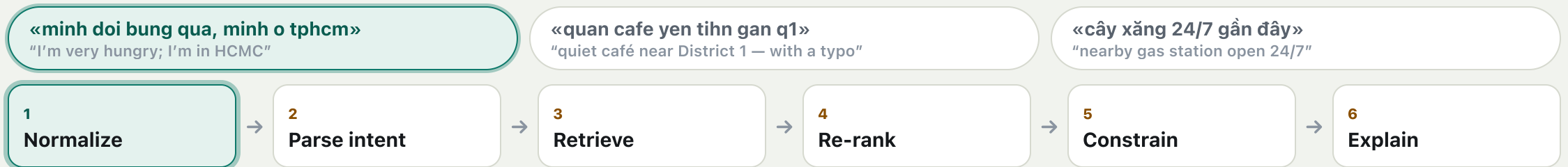
1 **Lẩu Phở Quận 1** Restaurant · HCMC

2 **Pizza 4P's Lê Thánh Tôn** Restaurant · HCMC

3 **Lẩu Phở Bình Thạnh** Restaurant · HCMC

Keyword search fails exactly where the sponsor warned: missing diacritics, typos, abbreviations, and queries with no category term. The bar: understand meaning, rank with multiple signals, **and explain every result**.

One Vietnamese query, traced through the six-stage pipeline



STAGE 1 OF 6 Clean the text

local · deterministic

INPUT

minh doi bung qua, minh o tphcm

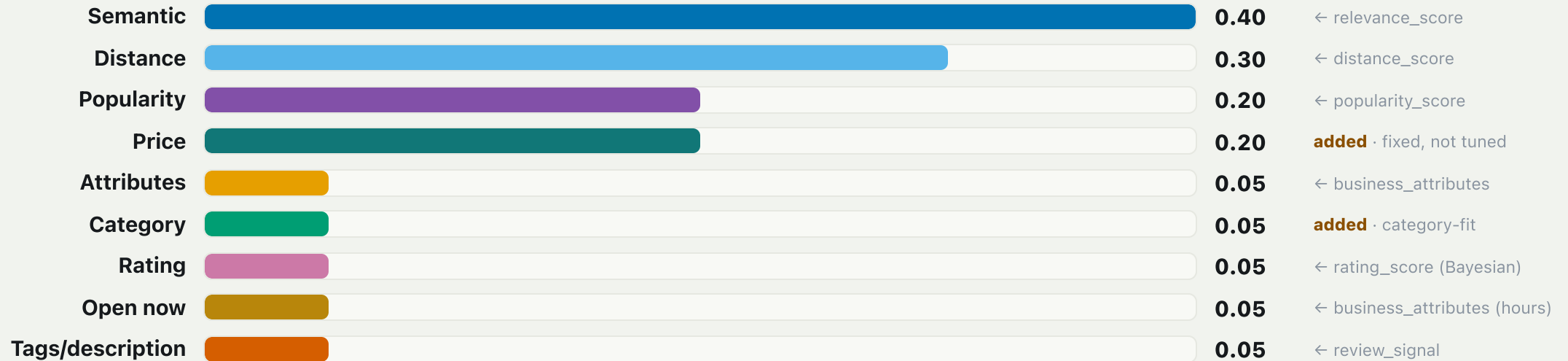


OUTPUT

minh đói bụng quá, mình ở TP.HCM

- **What happens:** Restore Vietnamese diacritics and expand "tphcm". The original query is still preserved in the API response.

9 signals, transparent weights — linear by design



6 signals map 1:1 to the sponsor's Ranking_Signals (business_attributes splits into attributes + open now), plus our 2 additions: **category-fit** and **price**. The dataset has no recency field for freshness_score, so we document it as roadmap rather than simulate it. Weights were tuned only on the tune split; **price is fixed at 0.20** (only 2/60 queries mention price — too few to tune). Explainable ranking is one of five official judging criteria, so we intentionally rejected a nonlinear model. After ranking: hard filters (location / entity / category) + never-empty relaxation.

Most hackathon teams do not measure — we do, with ablations

GATE	METRIC	RESULT	THRESHOLD	
G1	BM25 Recall@5 (tune)	0.929	≥ 0.55	✓
G2	hybrid NDCG@5 (tune)	0.933 > 0.881	> standalone	✓
G3	NDCG@5 / Recall@3 (test)	0.962 / 0.983	≥ 0.80 / 0.75	✓
G4	p95 latency (warm)	9.6 ms	< 200 ms	✓
G5	robustness (60 eval + adversarial)	138/138	0 failures	✓

Ablation (tune, NDCG@5): random 0.005 → BM25 0.877 → dense 0.881 → hybrid **0.933** → full re-rank **0.966** — every layer earns its place.

Method — guarding against self-deception

60 labeled queries → **40 tune / 20 test**, stratified by difficulty, with the seed committed to the repo. The test set is **evaluated exactly once per milestone** and never used to change code or weights. Test NDCG@5: 0.962; bootstrap 95% CI: 0.905–1.000.

9× corpus stress test — 1,000 POIs

Typo queries, NDCG@5: 0.510 (before) → **0.594** (local Damerau) → **0.737** (LLM rewrite). Warm p95 remains **57 ms** at 1,000 POIs. Also includes 150 synthetic queries with structured ground truth.

Field-by-field contract compatibility — and a demo that never dies

Tasco /v1/search — exact field compatibility

- PlaceResult: stable id poi:C001, WGS84, distanceMeters, score, tags
- ErrorResponse with correct error code + requestId
- Bearer / X-API-Key · X-Request-Id / X-Locale / X-Timezone
- openapi.json + **Dart adapter** → SearchSuggestion
- Vietnamese diacritics preserved in every output

Mode matrix — every layer has a fallback

MODE	EMBEDDINGS	LLM PARSE
local	bge-m3	off
local-first	local → Bedrock	off
cloud	Bedrock → BM25 fallback	on (auto-gated)

Network unavailable → local bge-m3. Everything unavailable → deterministic BM25 fallback. **The demo never dies.**

Built with AWS · deployment

- Bedrock cohere.embed-multilingual-v3 (embeddings)
- Claude Haiku 4.5 for parsing + query correction
- Railway: ~650 MB Docker image, ~5 s cold boot, /health reports its mode

Integrating into the Tasco app means **changing one base-URL line**. The extension endpoint /v1/semantic-search adds breakdown + reasons + intent — the schema-extension pattern allowed by the organizers.

Measured on untouched data — compatible with the published contract

0.962

NDCG@5 · TEST

0.983

RECALL@3 · TEST

9.6_{ms}

P95 (WARM)

• [demo link](#)

Thank you — try these Vietnamese queries:

quán cà phê yên tĩnh để ngồi làm việc
quiet café for working

minh đói bụng quá, mình ở tphcm
I'm very hungry; I'm in HCMC

nơi hẹn hò lãng mạn có view đẹp
romantic date spot with a beautiful view

cây xăng 24/7 gần đây
nearby 24/7 gas station